

Worksheet (Memo)

$$1. \lim_{x \rightarrow 0} \frac{\tan x}{x} = 0$$

$$2. \lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x} \quad (\text{Form } \frac{0}{0})$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \sec^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{-\tan^2 x} \quad (\text{Form } \frac{0}{0})$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{-2 \tan x \cdot \sec^2 x}$$

$$= \lim_{x \rightarrow 0} -\frac{1}{2} \cos^3 x$$

$$= -\frac{1}{2}$$

$$3. \lim_{x \rightarrow \infty} \frac{3 - e^{3x^2}}{3x - 6} \quad (\text{Form } \frac{-\infty}{\infty})$$

$$= \lim_{x \rightarrow \infty} \frac{-6xe^{3x^2}}{3} = -\infty$$

$$4. \lim_{x \rightarrow -\infty} \frac{e^{3x^2}}{3x} \quad (\text{Form } \frac{\infty}{-\infty})$$

$$= \lim_{x \rightarrow -\infty} \frac{6xe^{3x^2}}{3} = -\infty$$

$$5. \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} \quad (\text{Form } \frac{\infty}{\infty})$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$$

$$6. \lim_{x \rightarrow \pi/2} (\sec x - \tan x)$$

$$= \lim_{x \rightarrow \pi/2} \left(\frac{1 - \sin x}{\cos x} \right) \quad (\text{Form } \frac{0}{0})$$

$$= \lim_{x \rightarrow \pi/2} \frac{-\cos x}{-\sin x} = 0$$

$$7. \lim_{x \rightarrow \infty} (e^{2x} + 3x)^x \quad (\text{Form } \infty^\infty)$$

$$= \infty$$

$$8. \lim_{x \rightarrow \infty} x^3 e^{-x}$$

$$= \lim_{x \rightarrow \infty} \frac{x^3}{e^x} \quad (\text{Form } \frac{\infty}{\infty})$$

$$= \lim_{x \rightarrow \infty} \frac{3x^2}{e^x} = \dots$$

$$= \lim_{x \rightarrow \infty} \frac{6}{e^x} = 0$$

$$9. \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right) \quad (\infty - \infty)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^x - 1 - x}{x(e^x - 1)} \right) \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{e^x - 1}{(e^x - 1) + x(e^x)} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{e^x}{e^x + e^x + xe^x}$$

$$= \frac{1}{2}$$

$$10. \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{1}{x} \right) \quad (\infty \times 0)$$

$$= \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{x} \right)}{\frac{1}{x}} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{-\frac{1}{x^2}}{1 + \frac{1}{x}} / -\frac{1}{x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1$$

$$11. \lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{\ln x} \quad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{\ln x}} / \frac{1}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\ln x} = 0$$

$$12. \lim_{x \rightarrow \frac{\pi}{2}^-} (\sin x)^{\tan^2 x} \quad (1^\infty)$$

$$\text{let } y = (\sin x)^{\tan^2 x}$$

$$\Rightarrow \ln y = \tan^2 x \ln(\sin x)$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}^-} \ln y = \lim_{x \rightarrow \frac{\pi}{2}^-} \tan^2 x \ln(\sin x) \quad (\infty \cdot 0)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{\ln(\sin x)}{\cot^2 x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{\cos x}{\sin x} \cdot \frac{1}{-2 \cot x \operatorname{cosec}^2 x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^-} -\frac{\sin^2 x}{2}$$

$$= -\frac{1}{2}$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}^-} y = e^{-1/2}$$

$$13. \lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)^{\sin x} \quad (\infty^0)$$

$$\text{let } y = \left(\frac{1}{x}\right)^{\sin x}$$

$$\Rightarrow \ln y = \sin x \ln\left(\frac{1}{x}\right)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \ln y = -\lim_{x \rightarrow 0^+} \sin x \ln x \quad (0 \times \infty)$$

$$= -\lim_{x \rightarrow 0^+} \frac{\ln x}{\operatorname{cosec} x} \quad \left(\frac{-\infty}{\infty}\right)$$

$$= +\lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\operatorname{cosec} x \cot x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin x \sin x}{x \cos x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow 0^+} \frac{2 \sin x \cos x}{\cos x - x \sin x}$$

$$= 0$$

$$\lim_{x \rightarrow 0^+} y = e^0 = 1$$

$$14. \lim_{x \rightarrow \infty} (e^{2x} + 3x)^{1/x} \quad (\infty^0)$$

$$\text{let } y = (e^{2x} + 3x)^{1/x}$$

$$\Rightarrow \ln y = \frac{1}{x} \ln(e^{2x} + 3x)$$

$$\Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln(e^{2x} + 3x)}{x} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{2e^{2x} + 3}{e^{2x} + 3x} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{4e^{2x}}{2e^{2x} + 3} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow \infty} \frac{8e^{2x}}{4e^{2x}}$$

$$= 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = e^2$$

$$15. \lim_{x \rightarrow \infty} \frac{\sqrt{7+9x^2}}{1-2x}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 \left(\frac{7}{x^2} + 9\right)}}{x \left(\frac{1}{x} - 2\right)}$$

$$= \lim_{x \rightarrow \infty} \frac{|x| \sqrt{\frac{7}{x^2} + 9}}{x \left(\frac{1}{x} - 2\right)}$$

$$= \lim_{x \rightarrow -\infty} \frac{-x \sqrt{\frac{7}{x^2} + 9}}{x \left(\frac{1}{x} - 2\right)}$$

$$= -\frac{\sqrt{9}}{-2} = \frac{3}{2}$$

B 1. $f(x) = \frac{6}{4-2^x}$

A) The domain of f is $\mathbb{R} \setminus \{2\}$

B) $x = 0 \Rightarrow y = 2$

C) Not odd or even.

D) $\lim_{x \rightarrow \infty} \frac{6}{4-2^x} = 0$ and $\lim_{x \rightarrow -\infty} \frac{6}{4-2^x} = \frac{3}{2}$

Two HA at $y = 0$ and $y = \frac{3}{2}$ and.

$\lim_{x \rightarrow 2^-} \frac{6}{4-2^x} = \infty$ and $\lim_{x \rightarrow 2^+} \frac{6}{4-2^x} = -\infty$

V.A. at $x = 2$

E) $f'(x) = \frac{6(\ln 2)2^x}{(4-2^x)^2}$ which is never

zero but undefined if $x = 2 \Rightarrow$ this is the only critical value.

$$\frac{f' > 0}{\frac{1}{2}} \quad f' > 0$$

f increases on $(-\infty, 2) \cup (2, \infty)$

F) No local max/min exist.

G) $f''(x) = \frac{6(\ln 2)^2 2^x (4+2^x)}{(4-2^x)^3}$ which is never

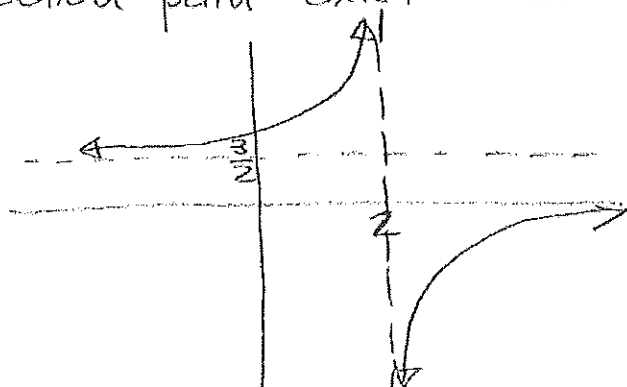
zero but undefined at $x = 2$

$$\frac{f'' > 0}{\frac{1}{2}} \quad f'' < 0$$

f is CU on $(-\infty, 2)$ and CD on $(2, +\infty)$

no inflection point exist since $f''(2)$ does not exist.

H)



$$2. \quad f(x) = x^3 - x^2 + 2$$

$$f'(x) = 3x^2 - 2x = x(3x - 2)$$

$f'(x) = 0$ at $x = 0$ and $x = \frac{2}{3}$ and exists everywhere $\Rightarrow$ critical numbers

$$x = 0 \text{ and } x = \frac{2}{3}$$

1st derivative test:

$$\begin{array}{ccccc} f' > 0 & & f' < 0 & & f' > 0 \\ & & \downarrow & & \downarrow \\ & & 0 & & \frac{2}{3} \end{array}$$

local max $f(0) = 2$ and local min $f(\frac{2}{3}) = \frac{50}{27}$

(P) 2nd derivative test:

$$f''(x) = 6x - 2$$

$$f''(0) = -2 < 0 \Rightarrow \text{local max at } x = 0$$

$$f''(\frac{2}{3}) = 2 > 0 \Rightarrow \text{local min at } x = \frac{2}{3}$$

$$3. \quad f(x) = x^4$$

$$f'(x) = 4x^3$$

$$f''(x) = 12x^2$$

$$\Rightarrow f''(x) = 0 \text{ if } x = 0 \text{ and the}$$

f'' is positive to the left and right of 0.

$$\begin{array}{ccc} f'' > 0 & & f'' > 0 \\ & & \downarrow \\ & & 0 \end{array}$$

The function f has no inflection point and is concave up on $(-\infty, 0) \cup (0, +\infty)$